

Restriction or misspecification? (Claude)

Dissecting classical and neural data assimilation

Do DA algorithms qualify estimation uncertainties? (Mine)

restriction — costs variance honestly, preserves calibration

misspecification — costs bias silently, destroys it

Data assimilation as Bayesian inference

$$\underbrace{p(x \mid C)}_{\text{posterior}} \propto \underbrace{p(y \mid x)}_{\substack{\text{observation likelihood} \\ H, R}} \times \underbrace{p(x \mid \theta, z)}_{\substack{\text{prior} \\ \text{dynamics } \Phi, \text{ model } M}}$$

State and context

x : the state trajectory over the window. $C = (y, \theta, z)$: observations over the window, per-window parameters θ , forcing z .

Prior from the dynamics, $p(x_0) \prod p(x_k \mid x_{(k-1)}, \theta, z)$; likelihood $\prod N(y_k; H x_k, R)$.

The key quantity: $E[x \mid C]$

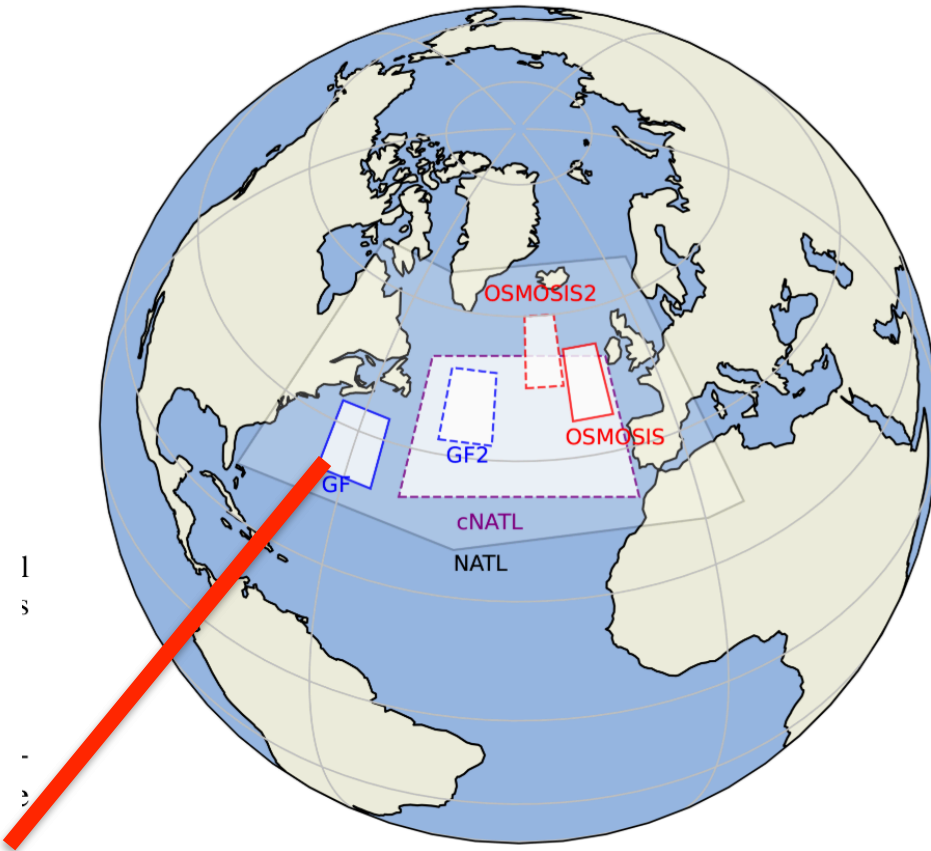
The conditional expectation is the MSE-optimal estimator. Its residual, $E \text{tr Cov}(x \mid C)$, is the irreducible error no scheme can beat.

DA algorithms for state estimation

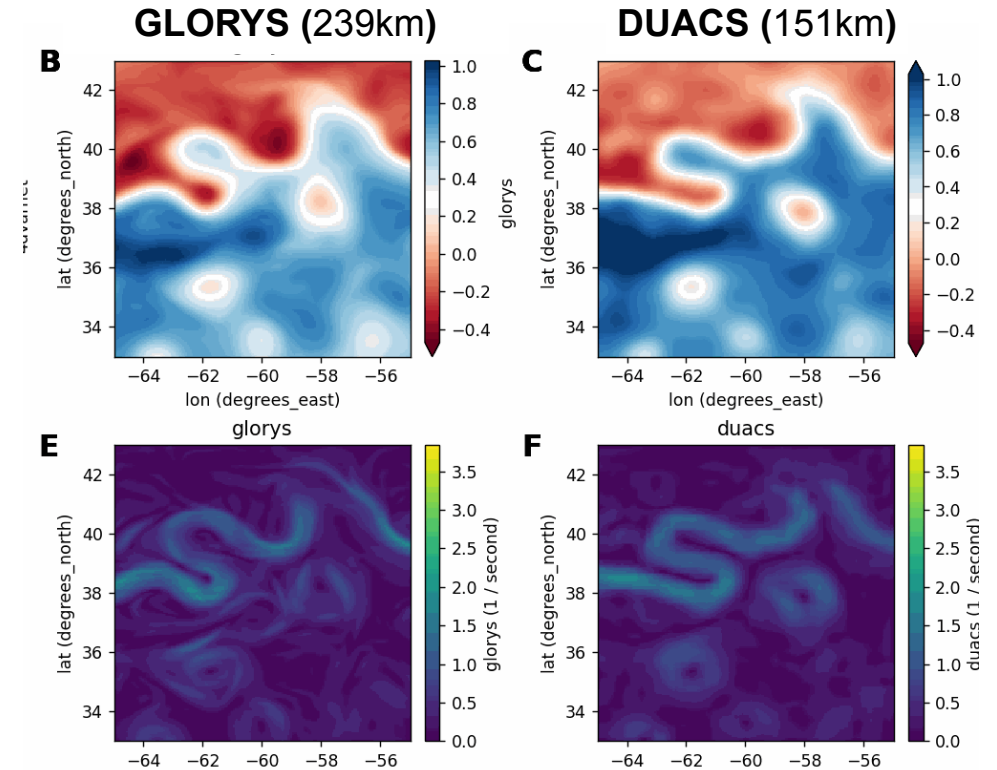
Sequential model-based DA algorithm : Markovianity + known dynamical model +

My questions

Why model-based DA underperforms compared with observation-based OI products?



An example for a Gulf Stream region

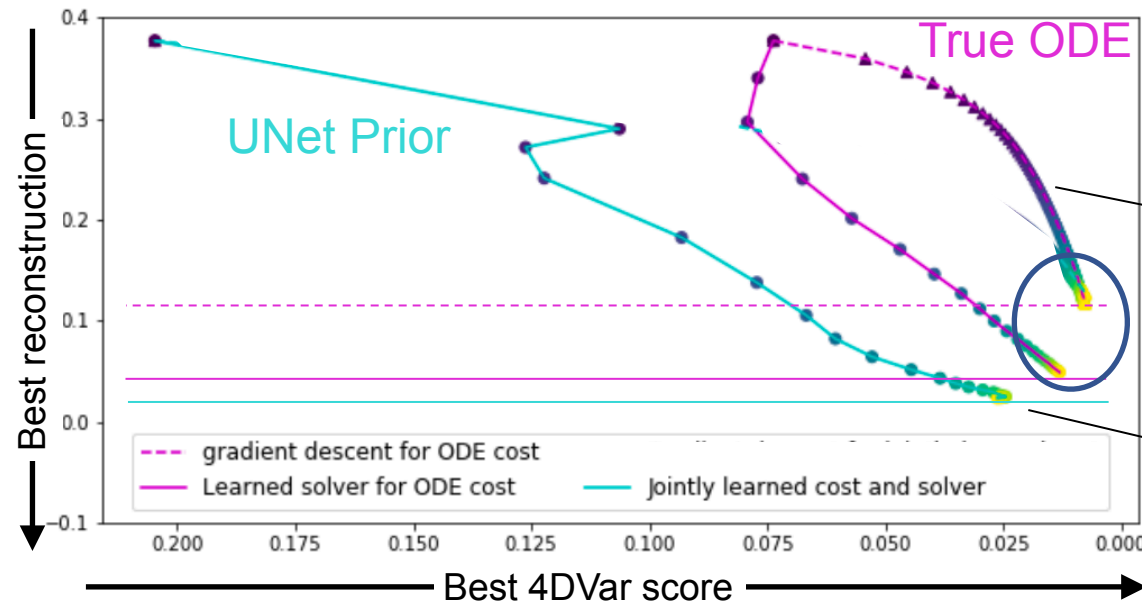


Sea Surface Height

Is this due to model errors? DA algorithms? Something else?
Many neural studies reporting better estimation performance than DA

My questions

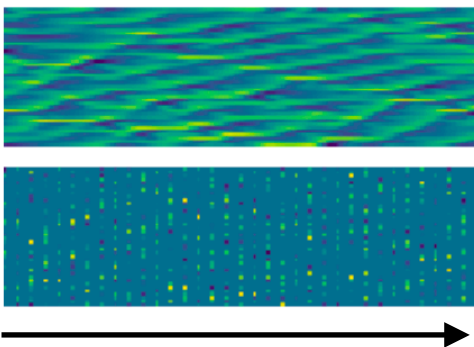
Why not using the true model in DA scheme can lead to better performance?



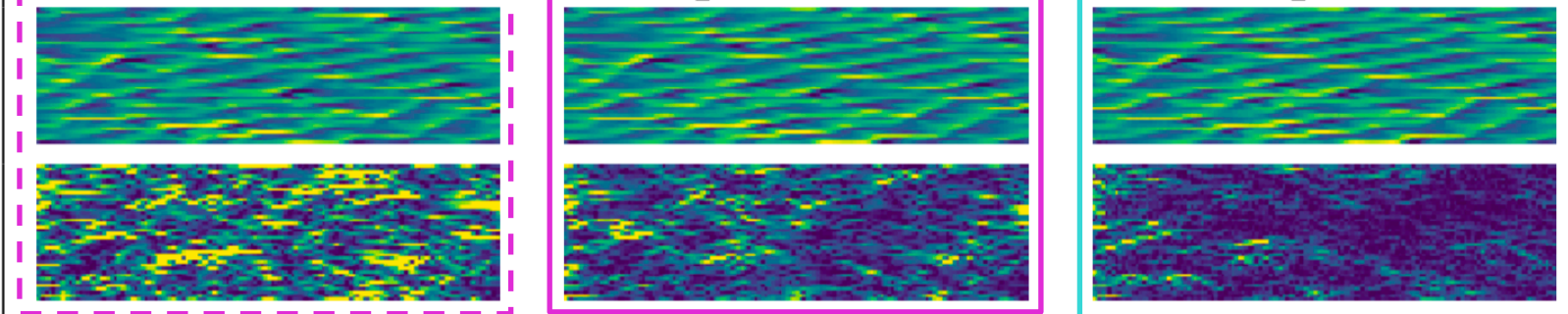
DA using the true ODE

Better MSE score not using the true prior

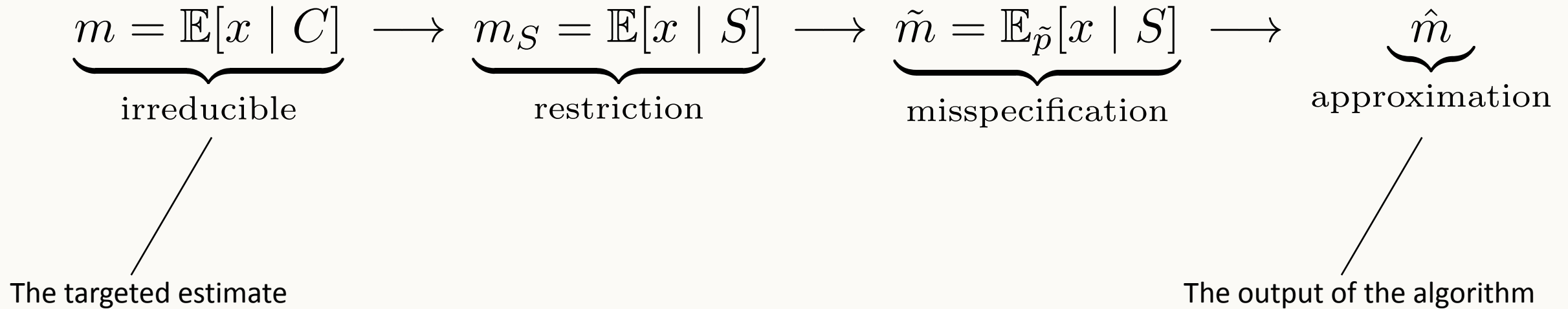
True and observed states



Reconstruction examples and associated error maps



Decomposition of the estimation error



$$\text{MSE} = \mathbb{E} \text{ Tr} [\text{Cov}(x \mid C)] + \mathbb{E} \|m - m_S\|^2 + \mathbb{E} \|m_S - \tilde{m}\|^2 + \mathbb{E} \|\tilde{m} - \hat{m}\|^2$$

Error chain of the conditional expectation

$$\text{MSE} = \underbrace{\mathbb{E} \text{Tr} [\text{Cov}(x \mid C)]}_{\text{irreducible}} + \underbrace{\mathbb{E} \|m - m_S\|^2}_{\text{restriction}} + \underbrace{\mathbb{E} \|m_S - \tilde{m}\|^2}_{\text{misspecification}} + \underbrace{\mathbb{E} \|\tilde{m} - \hat{m}\|^2}_{\text{approximation}}$$

DA filters vs. Smoothers

$C = \mathbf{y}_{t \in [0, T]}$ vs. $S = \mathbf{y}_{t' < t}$

MAP vs. Conditional expectation (eg, 4DVar)

Modelling errors (noisy parameters, missing model components,...)

Gaussian hypothesis,.....

Finite ensemble for the full pdf

Gradient descent for a global minimum

Finite training datasets and parametric architectures for NN

Do model-based vs. neural DA differ in the allocation of the error budget ?

Error chain of the conditional expectation: model-based vs. neural DA

$$\text{MSE} = \underbrace{\mathbb{E} \text{Tr} [\text{Cov}(x \mid C)]}_{\text{irreducible}} + \underbrace{\mathbb{E} \|m - m_S\|^2}_{\text{restriction}} + \underbrace{\mathbb{E} \|m_S - \tilde{m}\|^2}_{\text{misspecification}} + \underbrace{\mathbb{E} \|\tilde{m} - \hat{m}\|^2}_{\text{approximation}}$$

DA filter	> 0	$e^{\lambda\delta}$	> 0 (Finite Ensemble)
DA smoother	= 0	$e^{\lambda\delta}$	> 0 (Finite Ensemble)
End-to-end neural DA	= 0	= 0	> 0 (Finite Training set, Network architecture)

The error chain, scheme by scheme

Each scheme's estimate of $E[x | C]$, read along $m \rightarrow m_S \rightarrow \tilde{m} \rightarrow \hat{m}$. **S1**: what model error adds.

scheme	restriction	misspecification	approximation
EnKF (N = 30)	Filtering: x_k uses y up to k only.	Gaussian (linear) update; multiplicative inflation. S1: biased Φ and forcing in every forecast; inflation can't absorb a parameter error.	30 members for 40 dims, no localisation, perturbed-obs noise.
EnKS (ETKS, lag L)	Filtering removed up to lag L; residual beyond ($\approx 6 / 3$ cycles at $S0 / S1$).	+ linear-Gaussian backward cross-covariances; assumed reading of inflation. S1: the same bias, propagated backwards.	As EnKF, + transform products.
Strong-4DVar (6 × 500 steps)	Cycling: a sub-window ignores later obs, carried only by a static background.	Hard constraint ($Q = 0$); Gaussian static B; mode, not mean; anomaly = 0. S1: biased Φ as a hard constraint (floor ≈ 1.4).	Non-convex L-BFGS over 500 chaotic steps.
DirectUNet	Sees y only: ignores θ , z (marginalises the training spread).	Point output (anomaly = 0); amortised map, off-distribution under observing-system shift. S1: none — trained on the true system (oracle confound).	Finite network and training (1200 epochs).
CFM	Sees y only: ignores θ , z (marginalises the training spread).	None structural in distribution; amortised map, off-distribution under observing-system shift. S1: none (oracle confound).	Finite network and training (1200 epochs); ODE sampler, 30 members × 20 Euler steps.
SDA (Score-based DA) Bayesian sampler using a trained FM prior	Depending on the configuration SDA may ignore θ and z ;	the guidance only approximates $\nabla \log p(y x_t)$, which acts like a shrunk observation covariance R . Possible misspecification if using biased θ and z	Finite network and training (1200 epochs); ODE sampler, 30 members × 20 Euler steps.

My questions

How do DA schemes inform the irreducible uncertainty with error-free dynamics?

How does model misspecification impact the ability to inform this irreducible uncertainty?

Does the form of the considered prior drive the performance ?

Experimental setup: S0 and S1

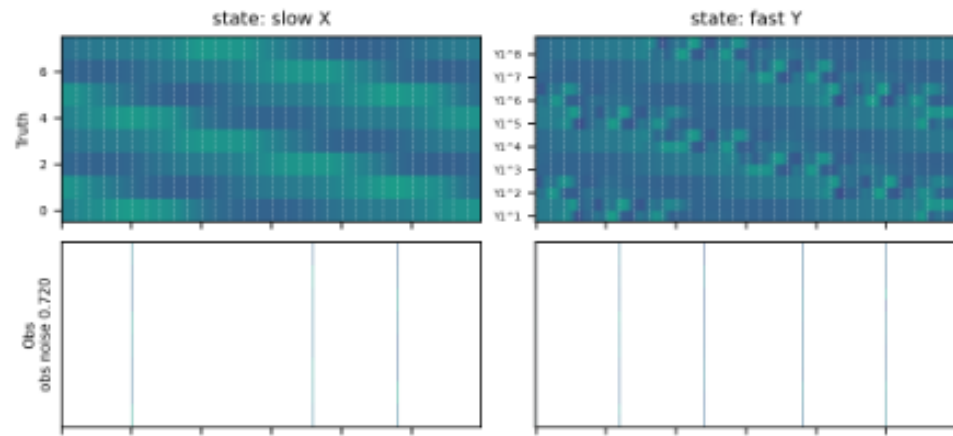
Two-scale Lorenz-96

State: 8 slow + 32 fast variables (40 dims); 3000-step windows ($dt = 0.001$).

Observations: 24 of 40 dims, noise variance $R = 0.5$; a random observation layout (10–100 times, 4–16 fast channels).

Context $C = (y, \theta, z)$: every window draws its own parameters θ (forcing, couplings, $\pm 20\%$) and a stochastic forcing z .

Test: 200 windows; every scheme sees the same y , noise and forcing.



S0 — PERFECT MODEL

The assimilating model runs with the window's true parameters θ^* and forcing. No dynamics error: only structural misspecifications (Gaussian update, hard constraint) remain.

S1 — MODEL ERROR

The model is told biased parameters θ_{da} (+10%) and a corrupted forcing. The truth is the same as at S0; only what the scheme is told changes.

S1 prices model error only for schemes that consume model information. Learned rows trained on the true system are a reference bound (the oracle confound).

How do DA schemes inform the irreducible uncertainty with error-free dynamics?

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The assimilating model runs with the window's true parameters θ^* and forcing. No dynamics error: only structural misspecifications (Gaussian update, hard constraint) remain.

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DA filter

0.7-0.8

<~30%

DA smoother

0.57-0.60

0

~30%

**End-to-end
neural DA**

0.44-0.46

?

0%

?

How does model misspecification impact the ability to inform this irreducible uncertainty?

S1 — MODEL ERROR

The model is told biased parameters θ_{da} (+10%) and a corrupted forcing. The truth is the same as at S0; only what the scheme is told changes.

$$\text{MSE} = \underbrace{\mathbb{E} \text{Tr} [\text{Cov}(x \mid C)]}_{\text{irreducible}} + \underbrace{\mathbb{E} \|m - m_S\|^2}_{\text{restriction}} + \underbrace{\mathbb{E} \|m_S - \tilde{m}\|^2}_{\text{misspecification}} + \underbrace{\mathbb{E} \|\tilde{m} - \hat{m}\|^2}_{\text{approximation}}$$

DA filter

1.40-1.50 (0.7-0.8)

~30%

> ~200%

< ~15%

DA smoother

1.30-1.40 (0.57-0.60)

0

> ~200%

< ~30%

**End-to-end
neural DA**

0.45-0.47 (0.44-0.46)

?

0%

?

Does the form of the considered prior drive the performance ?

$$\underset{\text{posterior}}{p(x \mid C)} \propto \underset{\substack{\text{observation likelihood} \\ H, R}}{p(y \mid x)} \times \underset{\substack{\text{prior} \\ \text{dynamics } \Phi, \text{ model } M}}{p(x \mid \theta, z)}$$

Model-based DA

$$p(x \mid \theta, z) \text{ derived from the model (ODE)}$$

Score-based DA (Rozet and Louppe, 2023)

$$p(x \mid \theta, z) \text{ Trained from data (here flow matching)}$$

Does the form of the considered prior drive the performance ?

$$\text{MSE} = \underbrace{\mathbb{E} \text{Tr} [\text{Cov}(x \mid C)]}_{\text{irreducible}} + \underbrace{\mathbb{E} \|m - m_S\|^2}_{\text{restriction}} + \underbrace{\mathbb{E} \|m_S - \tilde{m}\|^2}_{\text{misspecification}} + \underbrace{\mathbb{E} \|\tilde{m} - \hat{m}\|^2}_{\text{approximation}}$$

DA filter

1.3-1.4 / 0.7-0.8

~30%

> ~200%

< ~15%

DA smoother

1.3-1.4 / 0.57-0.60

0

> ~200%

< ~30%

SDA wo forcing

0.61-0.62 / 0.61-0.62

< 1%

~30%

?

SDA w forcing

0.61-0.62 / 0.61-0.62

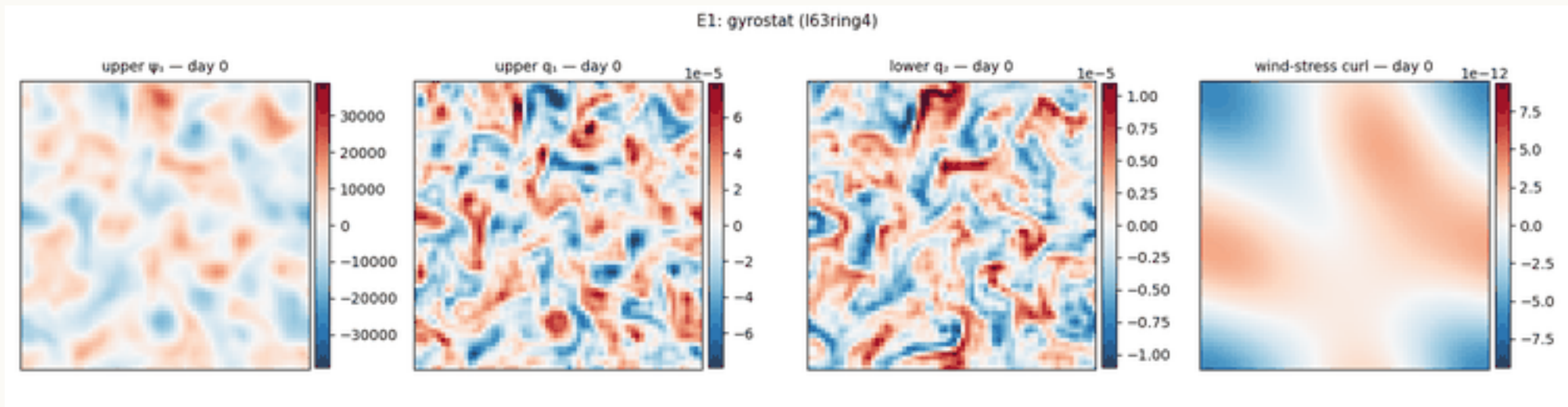
?

~30%

?

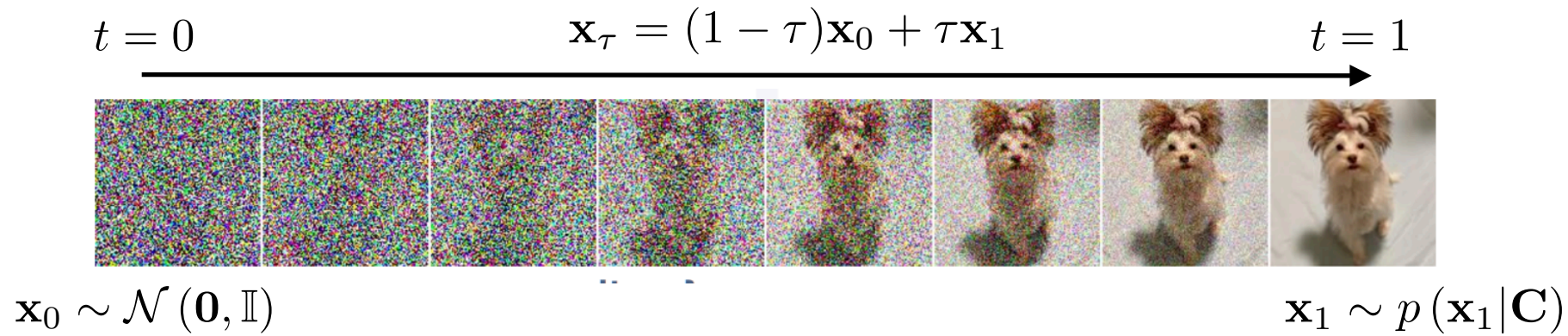
Synthesis

- Decomposition of the estimation error beyond the irreducible uncertainty
- Improving the prior does not necessarily translate in improved state estimate
- Approximate priors may lead to better performance than the true physics
- Model-based DA may not inform properly on the actual information gain of conditioning variables (observations, model parameters, forcings)
- Coded using **opencode** and **Claude-code** <https://github.com/CIA-Oceanix/4dvarnet-fm-opencode>



What next? Conditional Flow matching as a generic representation of DA algorithms

Flow matching path from a Gaussian space to the targeted physical space



Conditional ODE:
$$\frac{d\mathbf{x}_\tau}{dt} = \frac{1}{1 - \tau} [\mathbb{E}[\mathbf{x}_1 \mid \mathbf{x}_\tau, \mathbf{C}] - \mathbf{x}_\tau]$$

Velocity operator decomposed according to mean and anomaly components :

$$\mathbb{E}[\mathbf{x}_1 \mid \mathbf{x}_\tau, \mathbf{C}] = \mathbb{E}[\mathbf{x}_1 \mid \mathbf{C}] + \Psi_{\text{anom}}(\mathbf{x}_\tau, \tau, C)$$

Flow matching as the measuring instrument

Any DA posterior — an ensemble, a point estimate, a learned sampler — is associated with a flow through its own conditional expectation $E_{\hat{p}}[x_1 | x_t, C] = \Psi_{\text{mean}} + \Psi_{\text{anom}}$. Every scheme becomes the same kind of object.

Every scheme is a flow

$\Psi_{\text{anom}} = 0$ — point estimates (4D-Var, DirectUNet): a degenerate flow.

Ψ_{anom} from the ensemble — filters and smoothers.

Ψ_{anom} learned — CFM, SDA.

Blending composes Ψ_{mean} from one scheme with Ψ_{anom} from another.

The error chain of the conditional expectation

estimator	conditioned on	term
$m = E[x C]$	true system, full context C	irreducible
$m_S = E[x S]$	true system, summary S of C	restriction
$\tilde{m} = E_{\tilde{p}}[x S]$	the scheme's own model $\tilde{p}$	misspecification
$\hat{m}$	what is actually computed	approximation

$$\underbrace{m = \mathbb{E}[x | C]}_{\text{irreducible}} \longrightarrow \underbrace{m_S = \mathbb{E}[x | S]}_{\text{restriction}} \longrightarrow \underbrace{\tilde{m} = \mathbb{E}_{\tilde{p}}[x | S]}_{\text{misspecification}} \longrightarrow \underbrace{\hat{m}}_{\text{approximation}}$$

Extending the considered decomposition to the full posterior distribution (beyond the)

Experimental design: matched information

Testbed

Two-scale L96: regular 30-obs grid, a canonical random layout, an $n_{\text{obs}} \times k$ factorial, and out-of-range probes.

Matched-information rule

Every main arm sees the same y and the same model information M — a biased θ_{da} , the DA forcing. Only the reference bounds (S_0 , learned-on-truth) see the true system.

Model-error axis

Parametric bias $\delta \in \{0, 5, 10, 20, 40\%\}$ crossed with stated uncertainty $\sigma \in \{0, 20\%\}$, plus one structural error.

Tuning parity

Every inference hyper-parameter (inflation, lag, Q , guidance weight, τ_o) is tuned on validation windows.

Results A — a perfect model: what each choice costs

Pooled calibration (spread / RMSE, pooled over windows × time × channels); calibrated ≈ 0.984 for N = 30 members.

scheme class	pooled S0	pooled S1	reading
Filters (ETKF, EnKF)	1.08–1.13	0.33–0.37	over-dispersed at S0, then the misspecification signature at S1 (86–89% unexplained)
Amortised flows (CFM, F1=no)	0.85–0.94	0.84–0.96	close to calibrated, invariant to model error
Factorised learned (SDA, hybrid; F1=yes)	0.52–0.76	0.52–0.91	genuinely under-dispersed, worse on random layout, invariant to model error

What F1, F2 and F3 buy

F3 — SEQUENTIAL INFERENCE

EnKF vs EnKS differ only in restriction — the F3 identity test. Smoothing recovers part of the filtering deficit at a perfect model.

F1 — PRIOR-LIKELIHOOD SPLIT

Explicit H, R buys observing-system robustness ($\times 1.2$ in-distribution vs $\times 1.3+$ under shift) and out-of-range robustness. Amortisation buys in-distribution accuracy.

F2 — DYNAMICS STRUCTURE

A learned joint-window prior (SDA, 0.501) beats the mechanistic one-step prior (Strong-4DVar, 0.703) at a perfect model.

Results B — under model error

MARGINAL VALUE OF OBSERVATIONS

7.0x

collapse in Strong-4DVar's gain from 15 → 30 observations, once model error is present. It then stays flat at the error floor out to 1000 obs.

Calibration breaks where uncertainty is explicit through Φ : ETKF's pooled spread/RMSE falls from 1.08 to 0.37 ($S_0 \rightarrow S_1$). Implicit, amortised uncertainty stays invariant (flows $\leq 25\%$ unexplained at both).

U-model: representing uncertainty converts misspecification back into restriction, only while $\delta \leq \sigma$. SDA2 (clean- θ training) leaks bias into the prior; SDA3-fix (noisy- θ training) stays flat (hybrid 0.310→0.333 vs 0.312→0.307).

Eight claims

#	claim	column	status
K1	Filtering is a restriction; smoothing recovers part of the deficit	F3	 pending
K2	Explicit H, R buys robustness; amortisation buys accuracy	F1	✓ done
K3	A learned window prior beats the mechanistic one-step prior	F2	🕒 partial
K4	Model error turns the hard constraint into a misspecification (headline)	U-model × F2	🕒 partial
K5	Representing uncertainty converts misspecification into restriction, while $\delta \leq \sigma$	U-model	🕒 partial
K6	Explicit uncertainty is tunable but fragile; implicit is near-calibrated and invariant	U-state × F1	✓ done
K7	Oracle gap: the value of true-system data	source	 pending
K8	Best scheme composes an amortised mean with a factorised, explicit-R sampler	F1 × U-state	✓ done

What classical DA still gets right — and where this study is limited

Classical DA's strengths

- The sparse-observation regime
- Observing-system transfer (F1: explicit H, R)
- Calibration, when its assumptions hold
- No training set required

Limits of the study

- One idealised system (a QG transfer table if needed)
- Circularity, only partly measured (§6.4)
- Tuning parity across every scheme
- No true posterior — differences are measurable, absolute terms bounded by the best scheme

What's left before submission

run	unblocks	status
ETKS smoother	§5.2, §5.3, §6.1	blocks submission
Weak-4DVar on L96	§5.3, §6.1, §6.2	blocks submission
SDA priors at 1200 epochs	§5.3, §5.4	blocks submission
M-trained arms ($\sigma = 0, 20\%$)	§6.1, §6.2, §6.4, §6.5	blocks submission
δ sweep + structural error	§6.1, §6.2	blocks submission
ETKF / EnKF rerun (post-#291)	§3, §5.5, §6.3	blocks submission

Part A (§5) is writable now except §5.2; Part B (§6) is where the pending runs land.

WORKING TITLE

What data assimilation's structural choices buy and cost: a controlled comparison with learned estimators under model error

SUPPLEMENTARY

Framing, thesis, and the design matrix

Why the design matrix replaces hypotheses, the restriction / misspecification thesis, the five commitments, and every scheme's row.

Why a design matrix, not hypotheses

What fell

Four hypotheses (H1–H3b), two symptoms, and the $\Psi_{\text{mean}} + \Psi_{\text{G}} + \Psi_{\text{NG}}$ partition as the measuring instrument.

C4 (impoverished posterior) is not supported.

The partition attributes nothing: $\text{DirectUNet} \approx \text{CFM}$. “ $\Psi_{\text{NG}} \equiv 0$ ” is not visible — and is not even true of the analysis ensemble.

What replaces it

A design matrix: every scheme is a point in (F1, F2, F3, U-state, U-model). Results are one-factor contrasts across it.

A two-component split $\Psi_{\text{mean}} + \Psi_{\text{anom}}$, plus an error decomposition: irreducible / restriction / misspecification / approximation.

Model error stays the headline — now a stress test applied across the matrix.

*“Every structural commitment of data assimilation is either an **information restriction**, which costs variance honestly and preserves calibration, or a **misspecification**, which costs bias silently and destroys it.”*

Model error turns commitments that are harmless restrictions under a perfect model into misspecifications. Representing model uncertainty converts the second back into the first — and each representation succeeds only over a range of model errors we can measure.

Five commitments, stated as choices

factor	question	options
F1	prior–likelihood split	factorised (H, R explicit) / amortised
F2	dynamics structure	one-step transition / joint window prior / none; source: equations vs samples
F3	sequential inference (old H1)	sequential / whole window
U-state	how state uncertainty is carried	none / explicit (B, P, ensemble) / implicit (learned)
U-model	how model uncertainty is represented (old H2)	none / state (Q, inflation) / parameters (θ spread) / structure

F3 is exact only if F2 holds in x and observation errors are white; model error breaks F2 in x , hence F3.

The design matrix: every scheme, one row

Table 1 of the paper — results become one-factor contrasts by reading across a column.

scheme	F1	F2	F3	U-state	U-model
ETKF / EnKF	yes	one-step Φ	yes	ensemble	inflation
Strong-4DVar	yes	Φ (hard)	cycled	static B	none
Weak-4DVar	yes	$\Phi + Q$	window	B, Q	Q
SDA1 / SDA2 / SDA3-fix	yes	learned joint	window	implicit + R	spread / θ -cond / noisy θ
DirectUNet	no	none	window	none	train spread
CFM (Vanilla, PredictState)	no	none	window	implicit	train spread
Hybrid DU $\rightarrow$ SDA3-fix	partly	joint window	window	implicit + R	noisy θ